BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

SECONDARY SCHOOL (10TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022 • Paper Code: Physics Paper–X (01) (22)

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** Physics Paper–X (01) (22)

۱۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. The bending of waves around the edges of the obstacle is:

(A) Refraction
(B) Diffraction
(C) Damping
(D) Reflection

Q2. A shock absorber is designed to introduce.....forces.

(A) Damping
(B) Electric
(C) Magnetic
(D) Gravitational

Q3. An average human ear can detect sounds with an intensity as low as.....W/m^2.

(A) 10^{-6}
(B) 10^{-8}
(C) 10^{-12}
(D) 1

Q4. The speed of sound on a warm day when the outdoor temperature is 38°C is:

(A) 331 m/s
(B) 345 m/s
(C) 351 m/s
(D) 354 m/s

Q5. The focal length of a convex mirror with radius of curvature 10 cm is:

(A) −5 cm
(B) +5 cm
(C) −10 cm
(D) +10 cm

Q6. The SI unit of electric field intensity is

(A) NC^2 (B) Nm

(C) NC^{-1} (D) NA

Q7. A material medium with high permittivity value.....the coulomb force between charges.

(A) Reduces (B) Increases

(C) Firstly increase then decrease (D) No effect

Q8. The device used to protect a circuit against overload is.....

(A) Heater (B) Rod

(C) Fuse (D) Lamp

Q9. When the current through a wire is 2.0 A and voltage across its ends is 3.0 V then resistance will be.....

(A) 1.5 Ω (B) 2.5 Ω

(C) 6 Ω (D) 4 Ω

Q10. The electronic circuit that gives low (0) output when all its input are high (1) is called.....gate.

(A) AND (B) NAND

(C) OR (D) NOR

Q11. Software that allows users to interact with the computer system is:

(A) Hardware (B) Application

(C) Operating system (D) Program

Q12. What type of nuclear decay leaves the number of protons and neutrons unchanged?

(A) Alpha decay (B) Beta decay

(C) Gamma decay (D) Both Alpha and Beta decay

Section-B

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 32

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 45 minutes.

QII. Answer any EIGHT parts. Each part carries FOUR marks.

1. Prove the relation between wave speed, wave length and frequency of wave.

2. In which medium air or water, an echo is heard sooner and why?

3. Light travels from crown glass (n_g = 1.52) into water (n_w = 1.33). The angle of incidence in crown glass is 40.0°. What is the angle of refraction in water?

4. Can we achieve total internal reflection from optically rare medium to optically dense medium?

5. Differentiate between direct and alternating current.

6. Why circuit breakers, fuses and switches are installed to "live wire"?

7. How can a magnetic field be used to generate electric current?

8. What is email and internet? List four uses of internet in daily life.

9. A heating element on an electric range operating on 240 V has a resistance of 30.0 Ω. What current does it draw?

10. Differentiate between analogue and digital electronics?

11. What are radioisotopes? Explain three uses for various applications.

Section-C

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 21

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 45 minutes.

Q. Attempt any THREE questions. All questions carry equal marks.

QIII.

(a) [4] What is wave motion? How waves can be categorized?

(b) [3] The distance between successive crests in a series of water waves is 4.0 m and the crests travel 9.0 m in 4.5 s. What is the frequency of the waves?

QIV.

(a) [4] Derive thin lens equation.

(b) [3] What is capacitor? Define capacitance and its units.

QV.

(a) [4] Describe the phenomena of mutual induction.

(b) [3] If the mutual inductance of two coils is 3.5 mH and the current through primary coil changes from 0 A to 10 A in 0.03 s, how much emf is induced in the secondary coil?

QVI.

(a) [4] What is nuclear fission? Explain nuclear fission chain reaction.

(b) [3] Describe the operation of AND, OR logic gates by drawing their symbol and truth table.